NOT MEASUREMENT SENSITIVE

P-D-680B
INTERIM AMENDMENT 2 (AS)
16 May 1995
SUPERSEDING
INTERIM AMENDMENT 1 (AS)
20 December 1994

FEDERAL SPECIFICATION

# DRY CLEANING AND DEGREASING SOLVENT

This interim amendment was developed by the Naval Air Systems Command for the Department of the Navy, Department of Defense based upon currently available technical information.

Page 1

1.1: Delete and substitute:

"1.1 Scope. Dry cleaning and degreasing solvent consists of five types of petroleum distillates for dry cleaning fabrics and for degreasing painted or unpainted metal parts during equipment maintenance or repair. Some of the low residue types (Type IA or IIA) contain terpene solvents."

1.2: Delete and substitute:

"1.2 Classification. Dry cleaning and degreasing solvent shall be of the following types:

| | |
|---|---|
| Type I | Regular (Flash point ≥ 38°C) (Military Symbol SD-1) |
| Type IA | Low residue (Flash point ≥ 38°C) |
| Type II | High flash point (Flash point ≥ 60°C) (Military Symbol SD-2) |
| Type IIA | Low residue (Flash point ≥ 60°C) |
| Type III | Very high flash point (Flash point ≥ 93°C) with low odor (Military Symbol SD-3)" |

1.2.1: Delete and substitute:

"1.2.1 NATO classification.

| | |
|---|---|
| Type I | - S-752 |
| Type IA | - None |
| Type II | - S-753 |
| Type IIA | - None |
| Type III | - None |

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Page 2

2.2: Under ASTM Standards, add:

D 1901 - Relative Evaporation Time of Halogenated Organic Solvents and Their Admixtures
F 331 - Non-volatile Residue of Halogenated Solvent Extract from Aerospace Components (Using Rotary Flash Evaporator)
F 484 - Stress Crazing of Acrylic Plastics in Contact with Liquid or Semi-Liquid Compounds

Page 4

Table I: Delete Table I and subsitute new Table I attached as Page 3 of this amendment.

Page 5

3.2: After 1st sentence, add: "Types IA and IIA may also contain terpenes."

Page 6

Table II: Delete and substitute:

| CHARACTERISTIC | ASTM METHODS | OTHER METHODS |
|---|---|---|
| Flash point | D 56 | |
| Distillation | D 86 | |
| Kauri-butanol value | D 1133 | |
| Allowable constituents, % volume | | FED-STD-141, method 7356 |
| Apparent specific gravity | D 1298 | |
| Aniline point | D 611 | |
| Color | D 156 | |
| Odor | D 1296 | |
| Copper corrosion | D 130 | |
| Acidity | D 847 | |
| Doctor test | D 235 | |
| Vapor pressure | D 2879 | EPA 420.1 |
| Total phenol content | | |
| Viscosity | D 445 | |
| Non-volatile residue (Type IA and IIA only) | F 331 1/ | |
| Evaporation rate | D 1901 2/ | |
| Acrylic and polycarbonate compatibility | F 484 3/ | |
| Dielectric strength | D 877 | |



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## *Table I. Dry Cleaning and Degreasing Solvent Properties.

| CHARACTERISTICS | TYPE I | TYPE IA | TYPE II | TYPE IIA | TYPE III |
|---|---|---|---|---|---|
| Flash point, °C, min | 38.0(100°F) | 38.0(100°F) | 60.0(140°F) | 60.0(140°F) | 93.3(200°F) |
| Distillation, °C: | | | | | |
| Initial boiling pt., min | 149 | * 120 | 177 | * 150 | 220 |
| 50% recovered | Report | - | Report | - | Report |
| Dry point, °C, max | 208 | * 210 | 211 | * 230 | 295 |
| Aniline point, °C | 57 to 74 | - | 57 to 74 | - | 73 TO 89 |
| Kauri-Butanol value | 29 to 45 | * 37 minimum | 29 to 45 | * 37 minimum | 27 TO 45 |
| Allowable constituents, (percent by volume: 1/ | | | | | |
| (a) Solvent with olefinic or cyclo-olefinic unsaturation, max | 5 | 0.8 | 5 | 0.8 | 0.8 |
| (b) Aromatic compounds with eight or more carbon atoms, except ethyl benzene, max | 8 | 0.8 | 8 | 0.8 | 0.8 |
| (c) Total of ethylbenzene, toluene and branched chain ketones, max | 20 | 1 | 20 | 1 | 1 |
| (d) Total of (a)+(b)+(c), max | 20 | 1 | 20 | 1 | 1 |
| Total chlorine content (ppm), max | 100 | 100 | 100 | 100 | 100 |
| Total phenol content (ppm), max | 0.5 | 0.5 | 0.5 | 0.5 | 0.5 |
| Apparent specific gravity | 0.754 to 0.820 | * 0.75 to 0.95 | 0.754 to 0.820 | *0.75 to 0.95 | 0.754 to 0.820 |
| Non-volatile residue, (mg/100ml), max 2/ | 10 | 2.5 | 10 | 2.5 | 10 |
| Color, min | 25 | 30 | 25 | 30 | 30 |
| Odor 3/ | Characteristic & non-residual | Low & non residual | Characteristic & non-residual | Low & non residual | Low & non residual |
| Corrosion copper, max 4/ | 2A | 2A | 2A | 2A | 2A |
| Acidity | Neutral | Neutral | Neutral | Neutral | Neutral |
| Doctor test | Negative | Negative | Negative | Negative | Negative |
| Vapor pressure, Torr @ 20°C, max | - | 7 | - | 5 | 0.40 |
| Evaporation rate, min 6/ | - | 0.20 | - | 0.030 | - |
| Viscosity, cSt @ 25°C, max | - | - | - | - | 5.0 |
| Acrylic, polycarbonate compatibility 5/ | - | * Report | - | * Report | - |
| Dielectric strength, kV, max | - | 20.0 | - | - | - |

Footnotes to Table I:

1/ These maximum limits are as defined in Rule 102, South Coast Air Quality Management district regulations.
2/ For Types IA and IIA, non-volatile residues shall be detemined using ASTM Method F331 as modified in Table II.
3/ Samples of P-D-680 having satisfactory odor characteristics shall be used as reference standards.
4/ Test for three hours at 100°C.
* 5/ Use acrylic specimen A, loaded to 3000 psi and tested for 8 hours, and polycarbonate specimen (machined from MIL-P-83310 plastic) of the same dimensions, loaded to 2000 psi and tested for 2 hours
* 6/ Butylacetate = 1.0



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Footnote to Table II:

1/ Except that: a) maximum water bath temperature shall be 85°C and vacuum may be reduced to less than 5 mm Hg; b) acetone shall be used as rinse solvent to transfer residue; c) sample shall be dried to constant weight in an oven maintained at 105°C; and d) results shall be reported as mg/100 ml.2/ Except that the calculation shall be the "dry time" of the reference solvent divided by the "dry time" of the test material.3/ Using acrylic Specimen A.

Page 8

6.1: Delete and substitute:

"6.1 Intended use. These solvents are hydrocarbon solvents, some of which are used in the dry cleaning and coatings industry and most of which are used in the military as degreasers and cleaners for painted or unpainted metal parts."

Insert new paragraph after 6.1.1:

"6.1.1.1 Type IA. Type IA is a relatively fast evaporating low residue solvent compatible with acrylic and polycarbonate plastic, and having a high dielectric strength. Ventilation is required to prevent the accumulation of flammable solvent vapors."

Insert new paragraph after 6.1.2:

"6.1.2.1 Type IIA. Type IIA is a slower evaporating (than Type IA), low residue solvent. Ventilation is required to prevent the accumulation of combustible vapors."

6.3: Add to Table III:

| Type | Size/Unit of Issue | Military Symbol/ NATO Code | NSN | Dash Number |
|---|---|---|---|---|
| IA | 1 Gallon (Gl) | None | | 13 |
| IA | 5 Gallon (Cn) | None | | 14 |
| IA | 55 Gallon (Dr) | None | | 15 |
| IIA | 1 Gallon (Gl) | None | | 16 |
| IIA | 5 Gallon (Cn) | None | | 17 |
| IIA | 55 Gallon (Dr) | None | | 18 |

The margins of this amendment are marked with an asterisk to indicate where changes from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of



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this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Preparing Activity:
Navy - AS
Project No. 6850-N864



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